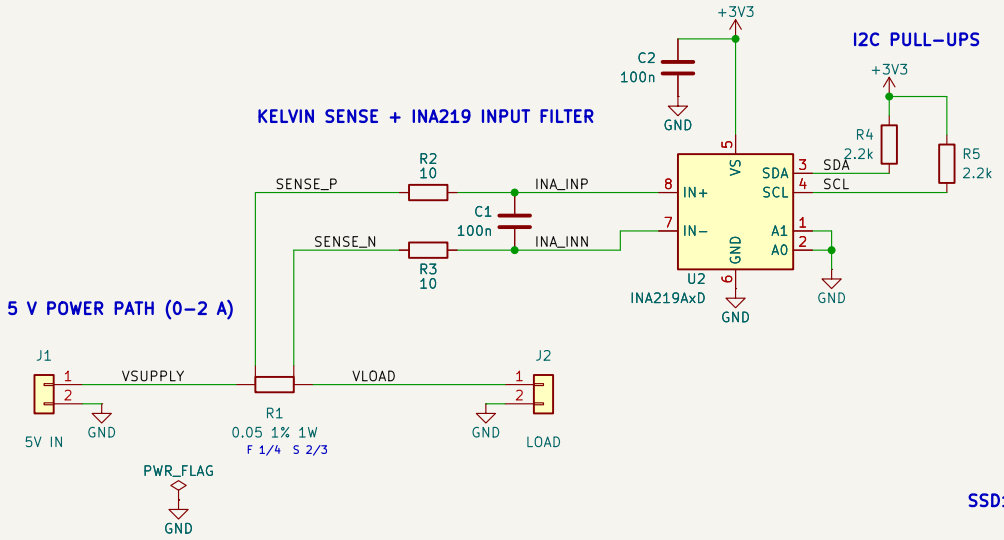
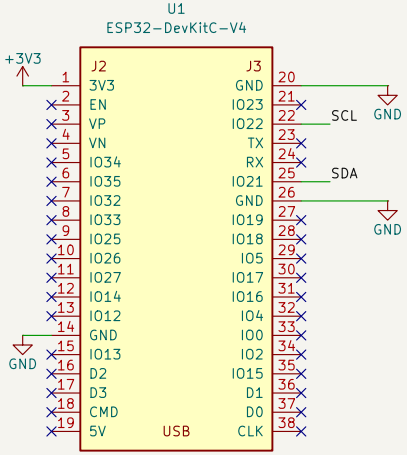
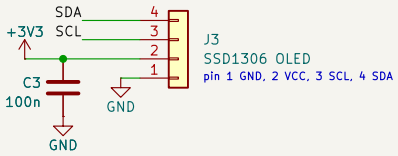






DESIGN NOTES
R1 shunt: 0.05 Ω, 1 %, 1 W, 2512, 4-terminal Kelvin (force pins 1–4, sense pins 2–3)
  V_shunt = I x R = 2 A x 0.05 Ω = 100 mV at full load (PGA /4 range ±160 mV -> 3.2 A full scale)
  P_shunt = I^2 x R = (2 A)^2 x 0.05 Ω = 0.2 W (5x margin on the 1 W part)
  Load voltage at 2 A: 5.00 V − 0.10 V = 4.90 V
Input filter: R2 = R3 = 10 Ω, C1 = 100 nF differential -> fc = 1 / (2π x 20 Ω x 100 nF) ≈ 80 kHz
INA219 (U2): CONFIG = 0x1777 (BRNG 16 V, PG /4, BADC = SADC = 64-sample average, shunt + bus continuous)
  CAL = trunc(0.04096 / (100 µA x 0.05 Ω)) = 8192; Current_LSB = 100 µA, Power_LSB = 2 mW
  Bus voltage is read at IN– (load side) against GND: load ground and logic ground are common
I2C: ESP32 GPIO21 = SDA, GPIO22 = SCL, 400 kHz, R4/R5 = 2.2 kΩ pull-ups to +3V3; U2 @ 0x40, OLED @ 0x3C
Power: U1 (ESP32-DevKitC V4) runs from its USB port; its 3V3 pin feeds U2 and the OLED; its 5V pin is unused

SSD1306 @0x3C, I2C 400 kHz, 2.2k pull-ups
INA219 @0x40 PGA/4 CAL=8192
Shon Zinger – EE portfolio

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Title: Smart Power-Monitoring IoT Node

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